

Department of Computer Science and Engineering

Academic Year 2020– 2021 (Odd Semester)

Degree, Semester & Branch: B.E, V, CSE

Course Code & Title: CS501 & Theory of Computation

Name of the Faculty member (s): Dr.I. Gethzi Ahila Poornima

Innovative Practice Description

- **Unit / Topic: Unit IV / Closure properties of Context Free Language**
- **Course Outcome:** CO4
- **Topic Learning Outcome: TLO1**
- **Activity Chosen: Reflection**
- **Justification:**

Understanding the concept of context free grammar is decided only by the level of awareness about the closure properties. Also, the closure properties are the base for further theorem proving. Hence determining the level of understanding of this concept is more important.

- **Time Allotted for the Activity: 25 Minutes**
- **About the Event**

Reflection is a process where teachers can identify their own teaching process and analyzing how their teaching (content delivery) might be improved for the better understanding of the students. The prime objective of Reflection is to make the students to discuss in team, write their doubts in a paper on topics which they are not clear, so as the teacher can discuss and clarify the concepts that are not understood by the students.

- **Details of the Implementation:**

At the end of the class, in order to speak out the doubts and to know the level of understanding of the students, they are insisted to honestly describe the concepts that they didn't understand well. The students wrote their doubts and topics which they are not clear. The papers are collected by the course instructor and discussed all the doubts in next class session and make them understand. The majority of the topic given by the students was one of the closure properties of CFL "Complement of CFL is not a CFL". Figure 1 shows sample sheets written by Ms. M. Harini and Ms.V. Vidhya.

- **CO – PO / PSO mapping:**

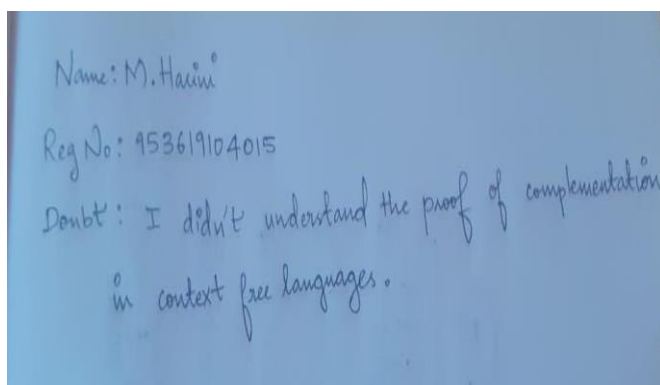
CO	PO1	PO4
CO2	1	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO4
	1	2
Justification for correlation	Students will be able to apply knowledge of mathematics to solve an engineering problem.	Students will be able to investigate the scope and importance of a problem using the Turing machine.

- **Images / Screenshot of the practice:**



NAME:- V. VIDHYA.
REG No:- 953619104048.
Doubt :- I didn't understand the proof of Concatenation in Context free Language.

Figure 1: Sample sheets

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- The students felt that this activity enabled them to have two-way communication and they will be able to understand it in the early stage.

- ❖ **Benefit of the practice:**

- Reflection helped me to identify the students and level of understanding, it helps to me to plan my next session accordingly.

- Students learned a few things individually that leads to self-learning.
They also actively involved in this session

❖ ***Challenges faced in implementation:***

- The reflection is conducted after the discussion of the topic and so there was shortage of time.

Signature of Faculty Member

HOD